

Aaliyah Phillips

10/9/2026

### Contribution Journal

For my contribution for the midterm, I was responsible for cleaning, prepping, and transforming the Telecom dataset, which specifically included encoding categorical variables, feature engineering, and one-hot encoding. In the Jupyter notebook, I made sure to label all my steps in my procedure as I worked. I address the missing values when needed, engineered several new features, converted categorical columns into numeric formats. Also, I showed a visual documenting the before and after of encoding and verified that there were no non-numeric columns in the dataset. I made sure to listen to my teammates' notes regarding how to go about the project and reflect that in my work before uploading to GitHub repository. Lastly I made sure to document the experience the my journals.

### Equine Ponce **Contribution Journal**

Elquin Ponce I completed feature scaling (StandardScaler) and normalization (MinMaxScaler) on the tenure, MonthlyCharges, and TotalCharges columns in the training data, including before and after distribution plots for both. Pushed this work to the shared GitHub repo. Class balancing (SMOTE) on the Churn target is pending completion of the encoding step from another section.

Seth Alvarez

Professor Rao

ITAI 1371

September 9, 2026

### Midterm Contribution Journal

For the midterm, I did the Exploratory Data Analysis section of the group notebook. I worked on the training data only, after Diane's 70/30 split. My section consists of nine steps: the

structure of the data, summary statistics for the numeric and categorical columns, the churn distribution, histograms of the numeric columns, box plots of the numeric columns by churn, churn rates for each categorical column, a correlation heatmap, and a check for missing values and duplicates.

I also wrote a summary at the end of my section to help plan the next steps. It covers the 7 missing `TotalCharges` values, the uneven spread of `TotalCharges`, the class imbalance, and which encoding fits each column. I let the group know that the cleaning and encoding sections had to go above the scaling cell, since that cell replaces the real values of `tenure`, `MonthlyCharges`, and `TotalCharges` in `train_data` with scaled ones. Anything that uses those columns after that cell would get the scaled numbers and not the original values.

## Contribution Journal: Data Cleaning

**Author:** Kei'Chara Flecks

**Section:** Data Cleaning (Missing Values, Duplicates & Data Types)

**Objective:** Standardize, clean, and verify the structural integrity of the training and testing datasets following the exploratory findings:

I Identified and resolved 7 missing `TotalCharges` values specifically linked to `tenure = 0` (brand-new accounts). I assigned mathematically sound values of `0.0` rather than relying on the mean/median, preserving initial relationship dynamics. Then isolated the duplicate removal process solely to the training set. The testing dataset was preserved with duplicates intact to emulate a realistic, unbiased environment. Programmatically enforced `float64` for charges, `int64` for integer metrics, and cast category attributes to stripped, sanitized strings to eliminate categorization issues caused by trailing spaces.

## **Contribution Journal Diane Lenhoff**

My contributions to the midterm included **researching possible datasets, working with the team to select the Telco Customer Churn dataset, supporting the professor's dataset approval, documenting the original dataset source, and completing and verifying the required Python 70/30 train/test split.**

I also **reviewed and integrated the team's notebook sections**, removed duplicate steps, corrected errors, and made sure each section flowed correctly into the next. I added the **Class Balancing section**, including before-and-after graphs, and ensured that only the training data was balanced while the test data remained unchanged. I also added the final export step to create the processed dataset.

Finally, I **prepared and organized all required documents needed to complete the midterm assignment**, including the Original Dataset URL document, one-page Dataset and Project Proposal, Contribution Journal, and Reflective Journal. I also organized the final GitHub repository to ensure all required files were included and clearly identified.